Lab 5

insert name here

2025-07-09

Short Answer Question

Question 1

Describe an example of unethical research in the medical field. Which of the four principles (Response for Persons, Beneficence, Justice, and/or Respect for Law and Public interest) does this violate, and why?

One unethetical research in the medical field was the Guatemala Syphilis Study where U.S researchers infected American soldiers, mental hospital patients, and others with syphilis and gave them pencilin to see how pencilin progresses.

Respect for Persons - no informed consent
Beneficence - giving people syphilis caused intentional harm rather than benefit the subject
Justice - aimed at the population in guatemala
Respect for Law + Public Interest - done secretly and did not obey Guatemala's laws and no one held accountability

Question 2

Describe an example of unethical research in the social science field. Which of the four principles (Response for Persons, Beneficence, Justice, and/or Respect for Law and Public interest) does this violate, and why?

Another example is the Stanford Prison Experiment (1971), where college students were assigned roles as guards or prisoners in a simulated prison. The study quickly escalated, and participants suffered emotional and psychological distress. This violated Beneficence because the researchers did not protect participants from harm. It also violated Respect for Persons, as participants were not fully informed of the potential risks and were not removed from harmful situations in time.

Question 3

Describe an example of unethical research in the digital-media field. Which of the four principles (Response for Persons, Beneficence, Justice, and/or Respect for Law and Public interest) does this violate, and why?

The Facebook emotional contagion study (2014), where researchers manipulated users' news feeds to see how it affected their emotions—without their informed consent. This violated the Respect of Persons since these subjects were not informed that they were being participated in this study. As well as about Beneficience since it risked harming users emotionally without any benefit.

Question 4

What's the difference between association and correlation?

Correlation is a specific type of association that measure a linear relationship between two numeric variables. +1 = perfect positive relationship, 0 = not linear, -1 = perfect negative.

Coding Question

Section 1: Set Up

Load `tidyverse`.

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

Load the data `storms`.

```r
data("storms")
```

Set your seed to a value of your choosing.

```r
set.seed(100)
```

Load the package `poliscidata`.

```r
library(poliscidata)
```

```
## Warning: package 'poliscidata' was built under R version 4.5.1

## Registered S3 method overwritten by 'gdata':
##   method         from
##   reorder.factor gplots
```

Load the data `states`.

```r
data("states")
```

Load the package `maps`.

```r
library(maps)
```

```
## Warning: package 'maps' was built under R version 4.5.1
```

```
##
## Attaching package: 'maps'
```

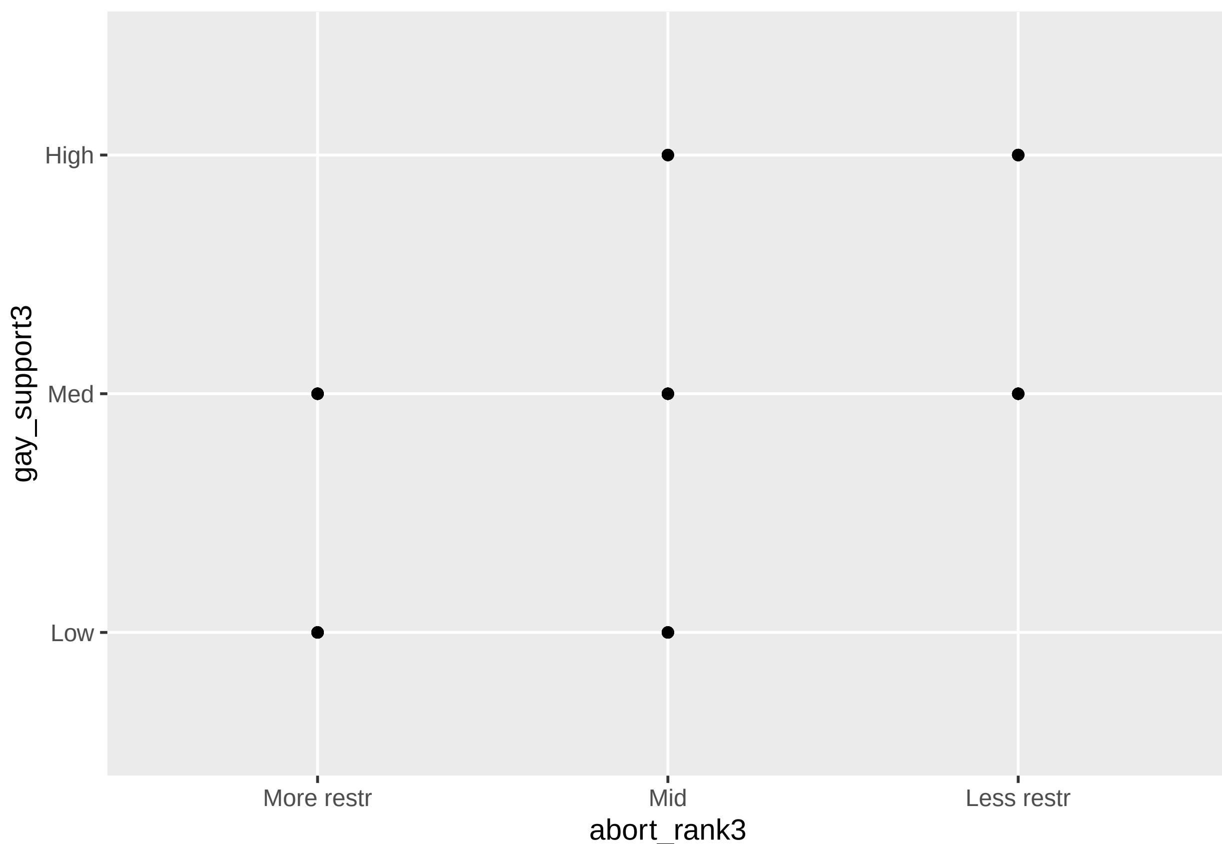
```
## The following object is masked from 'package:purrr':
##
##     map
```

Section 2: More with ggplot()

Task 1: Determine if there is a correlation between the abortion restrictions and public support for gay rights.

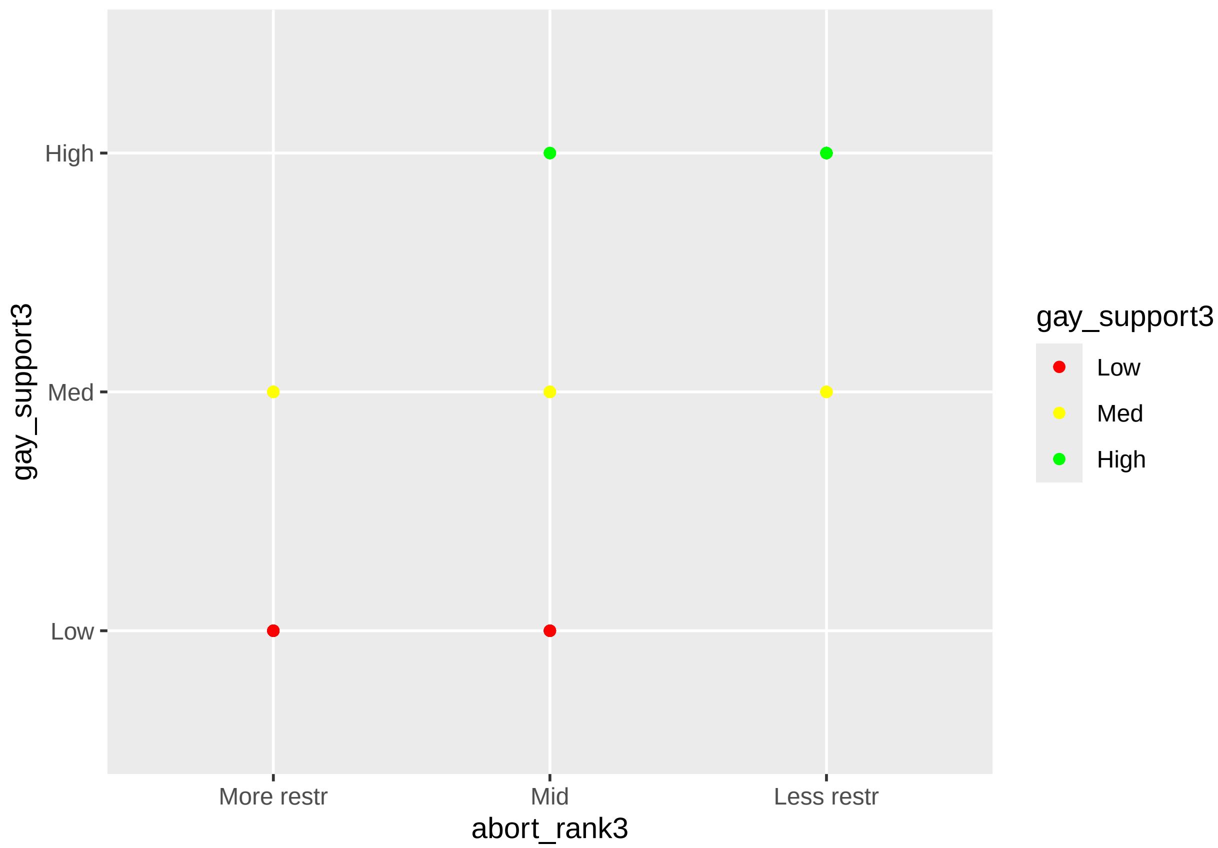
Create a plot that shows the association between Abortion restrictions (3 category ranking) and Gay rights: public supports (3 categories).

```r
states |>
  ggplot(aes(x=abort_rank3, y=gay_support3))+
    geom_point()
```



Change the colors so that low is red, medium is yellow and high is green.

```r
states |>
  ggplot(aes(x = abort_rank3, y = gay_support3, color = gay_support3)) +
     geom_point() +
     scale_color_manual(values = c("red", "yellow", "green"))
```



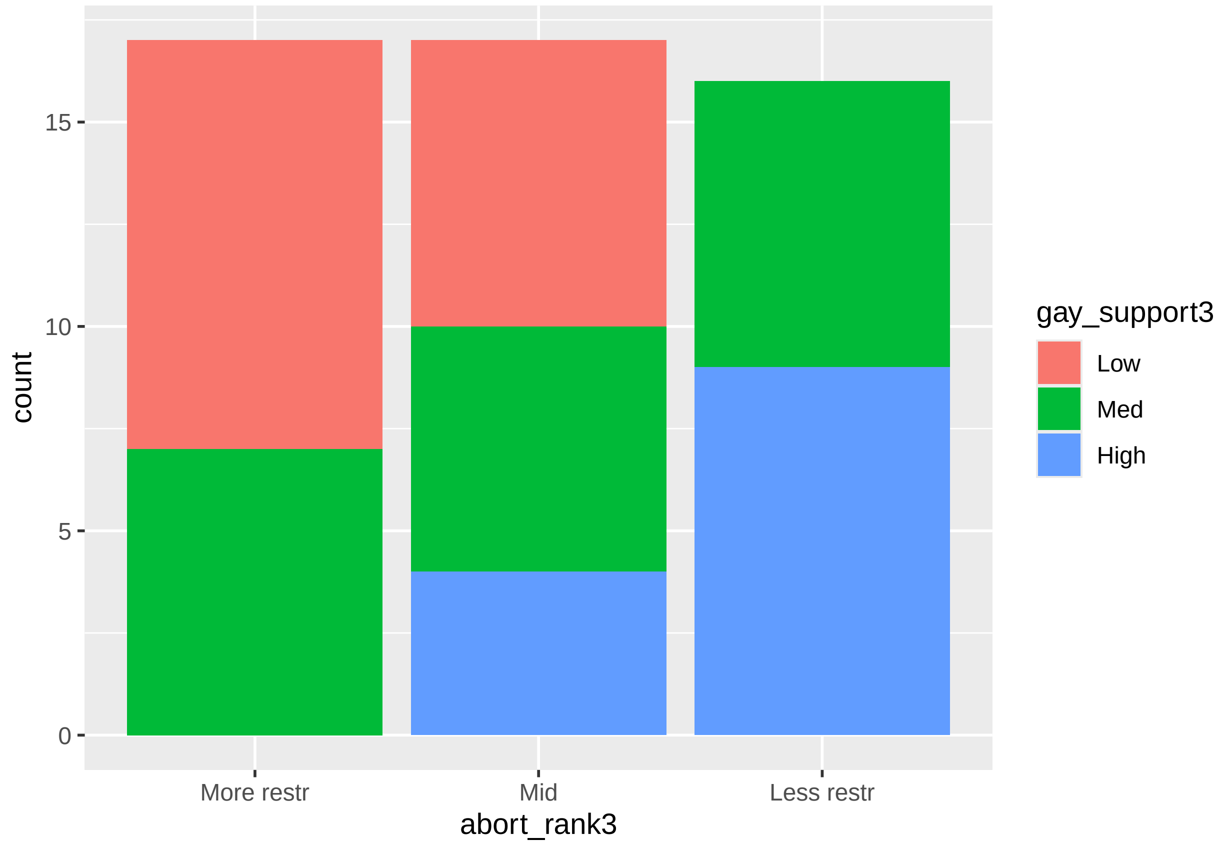
Is there an association between these two variables? How can you tell? And does this intuitively make sense?

There is some association since we can see how the less restrictive abortion rights has mid-high gay support compared to more restrictive abortion rights which had low to mid gay support.

Recreate the plot above without it being normalized (only a stacked bar chart). What new information does this show?

This shows how the stricter the policies the greater the gay support vice versa.

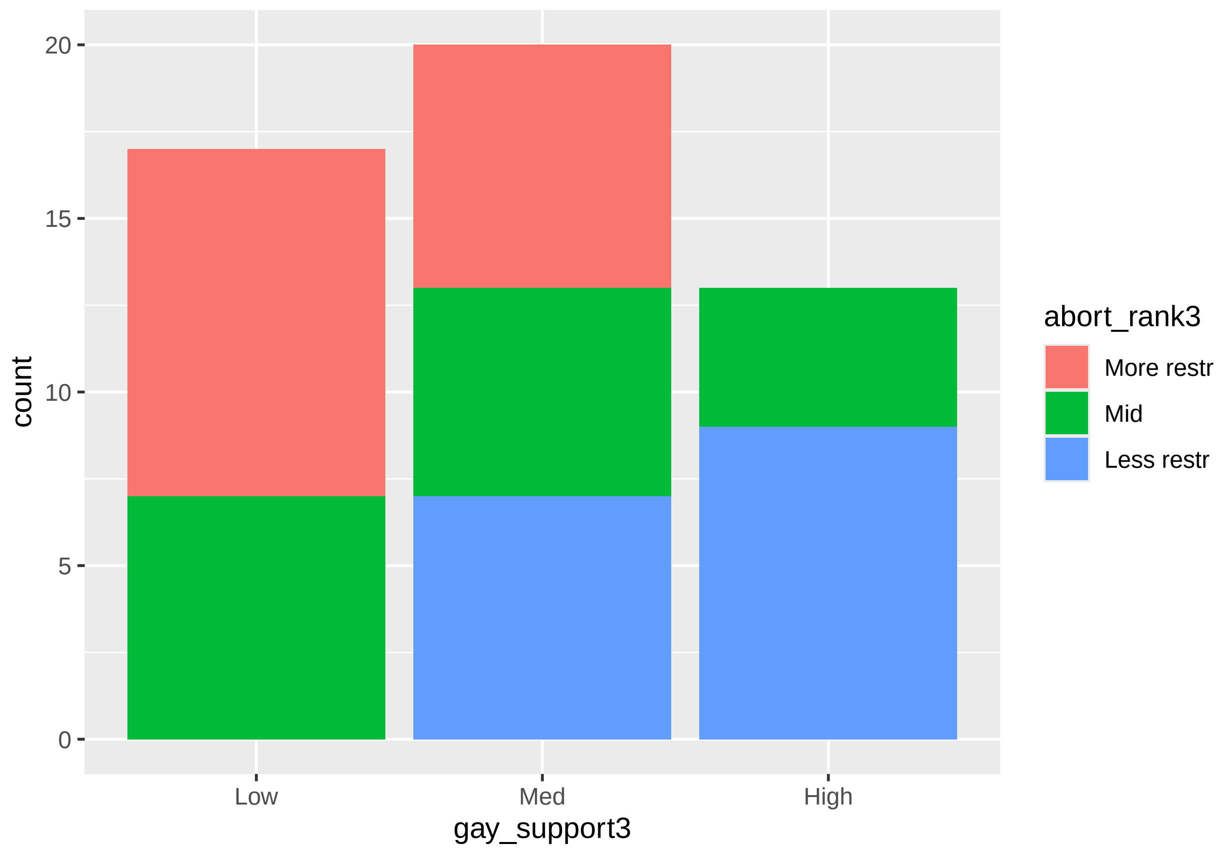
```r
states |>
  ggplot(aes(x=abort_rank3, fill=gay_support3))+
    geom_bar()
```



Switch the variable assignment of the aesthetics. What new information does this show?

There is no new information shown because the first graph shows how less restrictive has more gay suppport vice verse the second graph shows the same thing except now the x-y axises are interchanged.

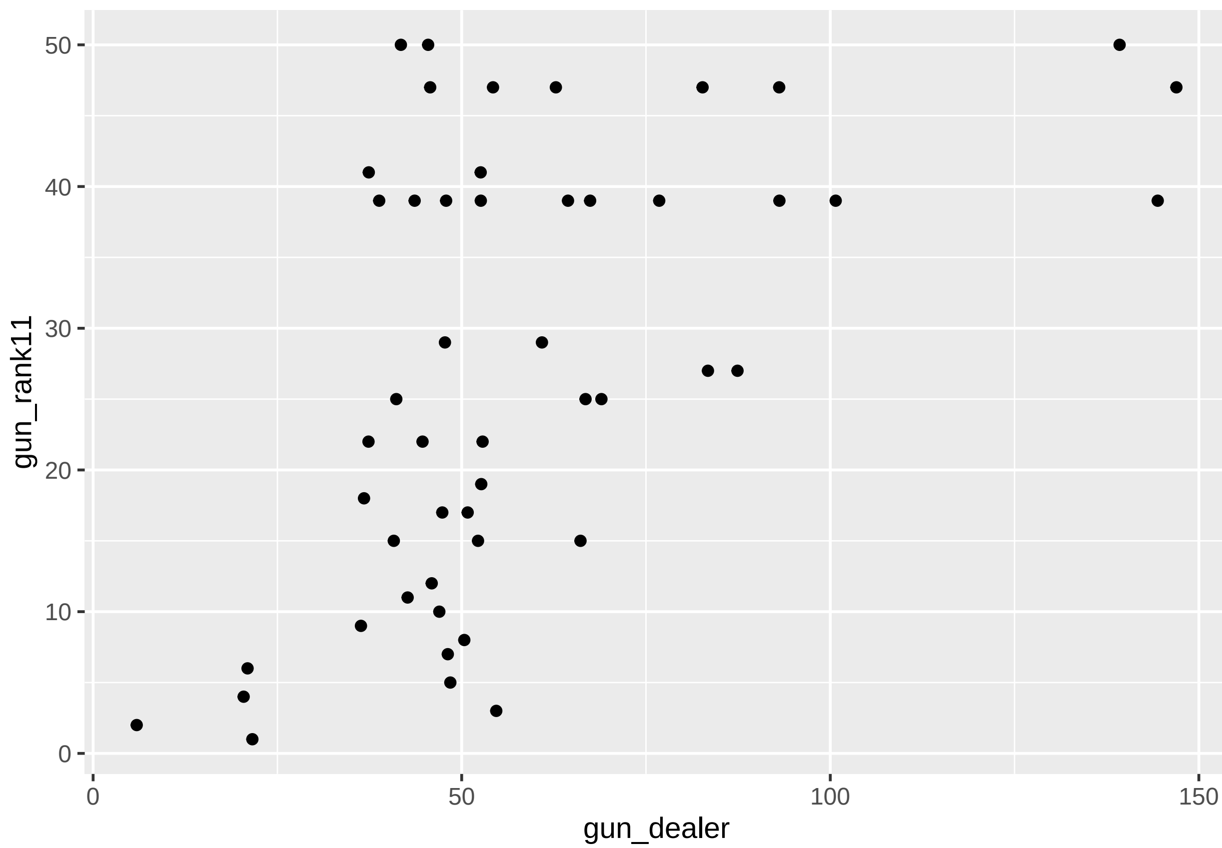
```
states |>
  ggplot(aes(x=gay_support3, fill=abort_rank3))+
    geom_bar()
```



Task 2: Plot two variables from the states dataset and determine if there is an association.

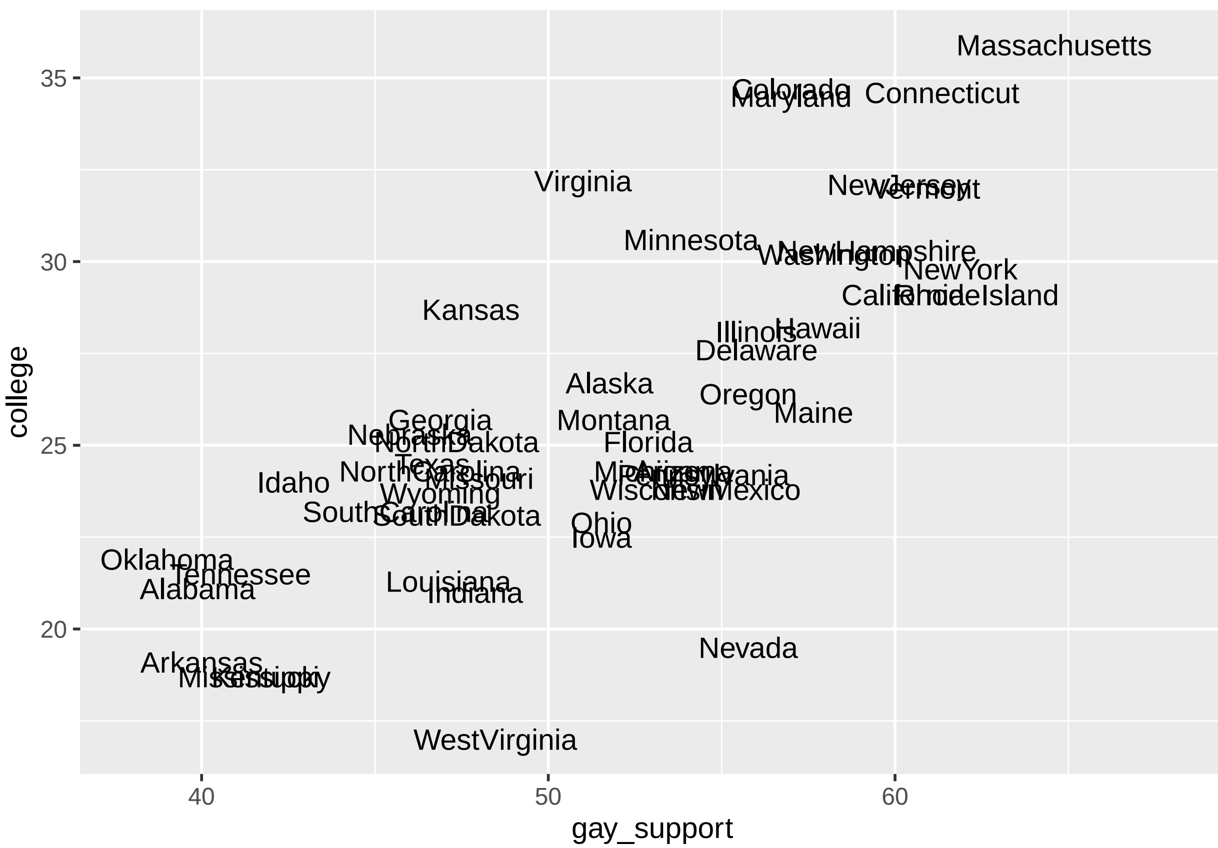
Written answer here.

```
states |>
  ggplot(aes(x=gun_dealer, y = gun_rank11))+
  geom_point()
```



Task 3: Use geom_text() to create a scatter plot of two variables from the `states` dataset.

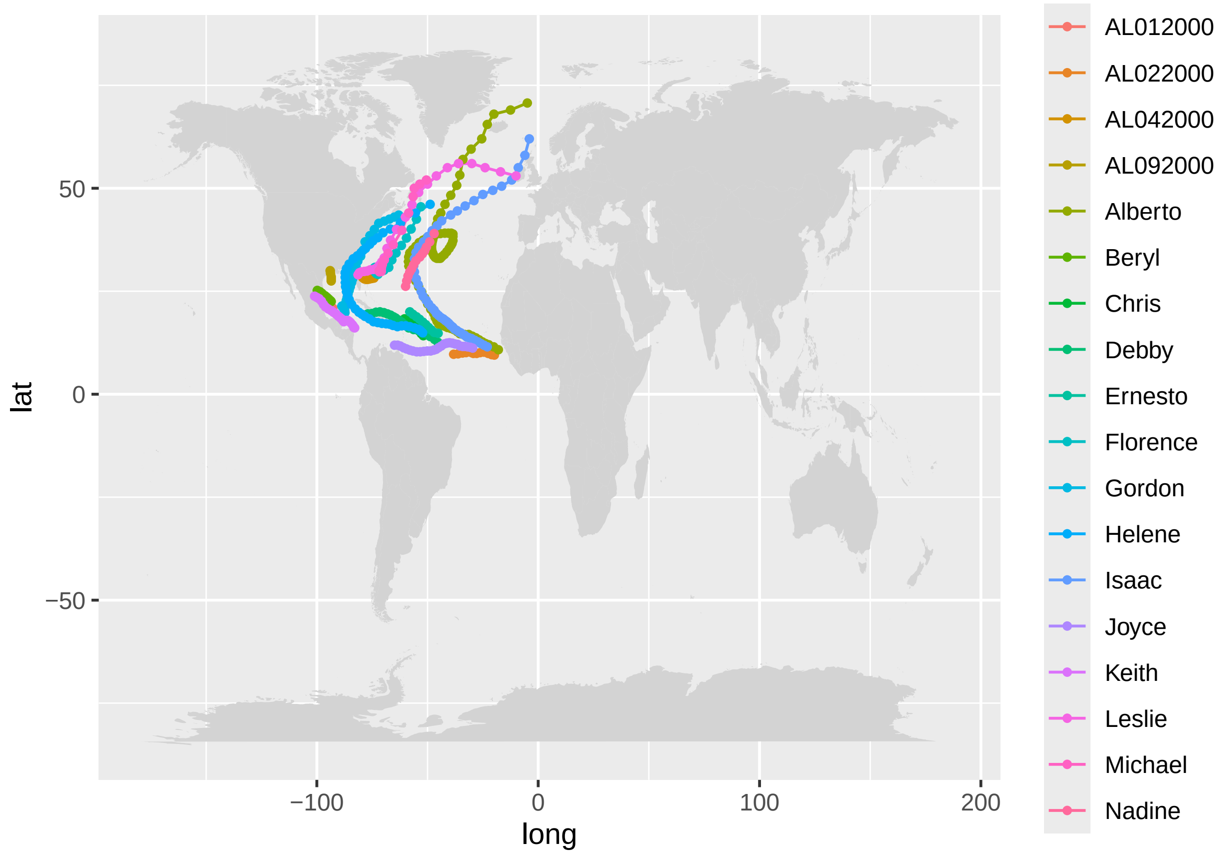
```
states |>
ggplot(aes(x=gay_support, y=college, label = state))+
  geom_text()
```



Section 3: Maps

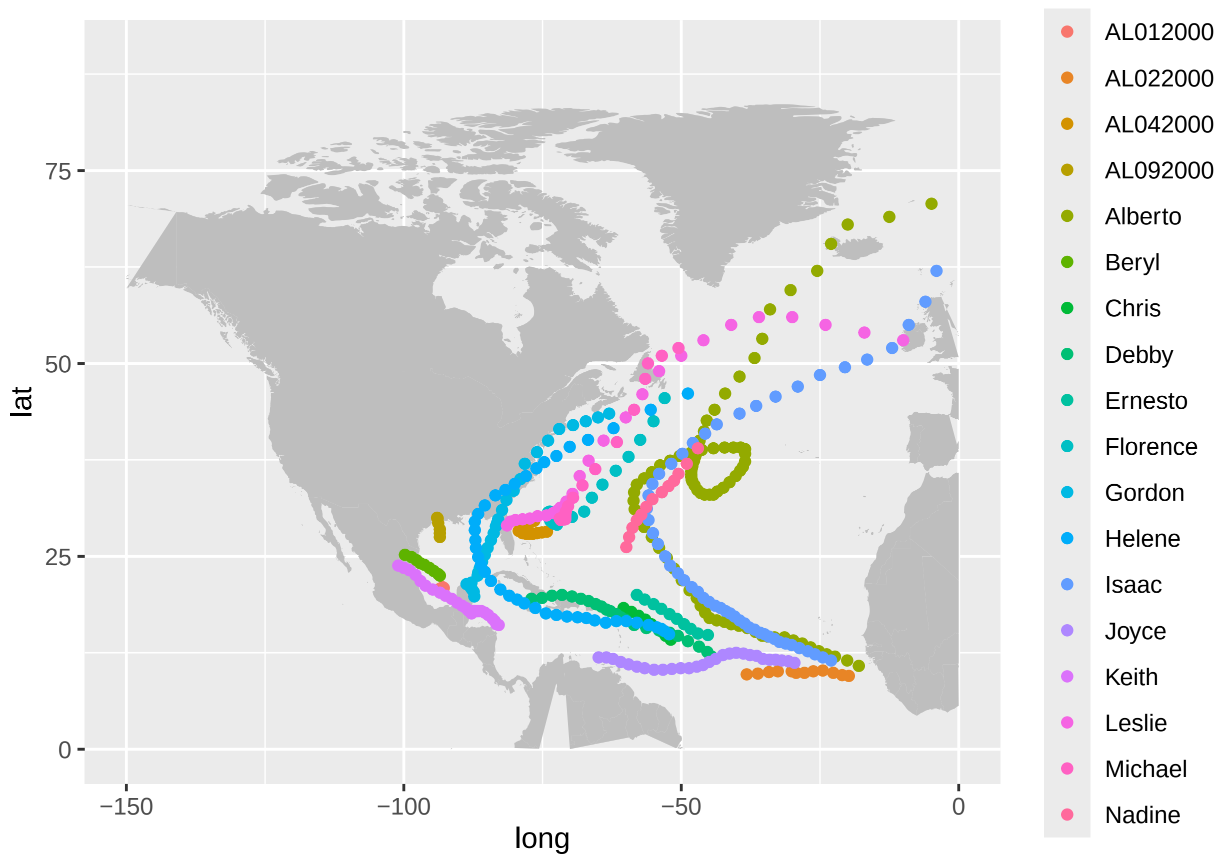
Task 1: Use the instruction here to map the `storms` in the year 2000, including longitude, latitude, name of storm and storm's maximum sustained wind speed, using `geom_path()`. Note that the example uses storms in '75.

```
storms_2000 <- storms |>
  filter(year == 2000)
world_map <- map_data("world")
ggplot() +
  geom_polygon(data = world_map, aes(x = long, y = lat, group = group),
               fill = "lightgray") +
  geom_path(data = storms_2000, aes(x = long, y = lat, group = name, color = name)) +
  geom_point(data = storms_2000, aes(x = long, y = lat, color = name), size = 1)
```



Task 2: Using similar code from above, try to create a gg_states object that plots the USA.

```
states_map <- map_data("state")
ggplot()+
  geom_polygon(data=world_map,
               aes(x=long,y=lat,group=group),
               fill="gray")+
  geom_point(data=storms_2000,
             aes(x=long,y=lat,color=name))+
  xlim(c(-150,0))+
  ylim(0,90)
```



Task 3: Join the `states` with `states_map`.

First, look at the formatting of the states names in either dataset. Why would this make a join difficult?

Hint: use `unique()` and `head()` .

Some of the states are written as South Dakota while the other dataset has it as southdakota so we will need to somehow make the South Dakota become southdakota by lowercasing all the characters of the string and removing any spaces.

```
unique(head(states$state, 10))
```

```
##  [1] Alaska
##  [2] Alabama
##  [3] Arkansas
##  [4] Arizona
##  [5] California
##  [6] Colorado
##  [7] Connecticut
##  [8] Delaware
##  [9] Florida
## [10] Georgia
## 50 Levels: Alabama                                ...
```

```
unique(head(states_map$region, 10))
```

```
## [1] "alabama"
```

Use `tolower()` and `gsub` to change all state names into all lowercase and no white spaces for BOTH datasets.

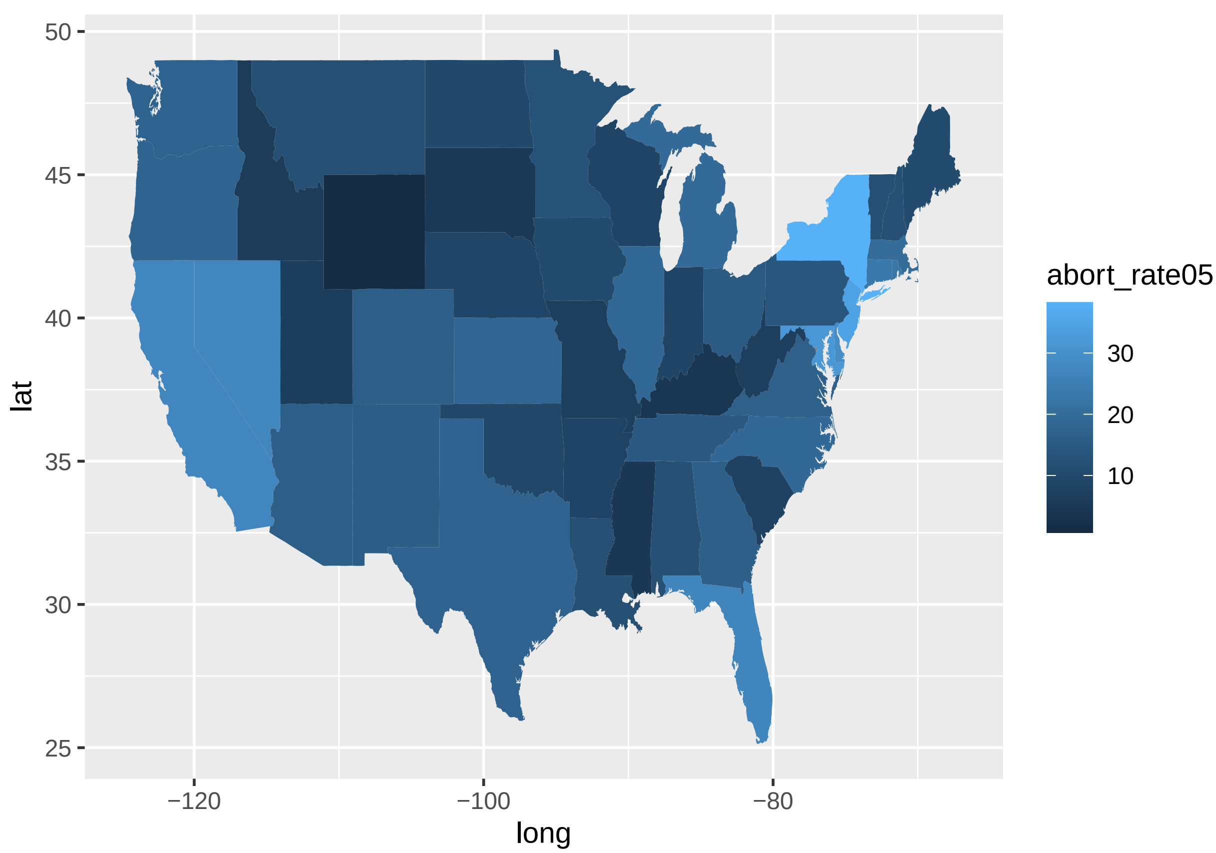
```
states <- states |>
  mutate(state = tolower(state),
  state = gsub(" ", "", state)
  )
states_map <- states_map |>
  mutate(region = gsub(" ", "", region))
```

Join the two datasets.

```
combined_states <- left_join(states_map, states, by = join_by(region == state))
```

Task 4: Use this joined dataset to plot an additional variable using fill.

```
ggplot()+
  geom_polygon(data = combined_states,
               aes(x=long, y=lat, group=group, fill=abort_rate05))
```



Section 4: Logistic Regression

Load the library `MASS` .

```r
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.5.1
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```r
data(crabs)
```

Use the index method from the KNN section of Lab 4 to create a test/train split of the `crabs` data.

```r
train_ind <- sample(1:nrow(crabs),
                    size=as.integer(.8*nrow(crabs)))
train <- crabs[train_ind, ]
test <- crabs[-train_ind, ]
```

Use `glm()` to fit a logistic regression for predicting species using frontal lobe size and carapace length.

```r
m1 <- glm(sp~FL+CL, family=binomial(link = "logit"), train)
m1
```

```
##
## Call:  glm(formula = sp ~ FL + CL, family = binomial(link = "logit"),
##     data = train)
##
## Coefficients:
## (Intercept)           FL           CL
##     -13.674       11.679       -5.185
##
## Degrees of Freedom: 159 Total (i.e. Null);  157 Residual
## Null Deviance:       221.7
## Residual Deviance: 38.9  AIC: 44.9
```

Save the predictions from this model into the training and testing datasets.

```r
train <- train |>
  mutate(predicted_prob = predict(m1, newdata = train, type = "response"))
test <- test |>
  mutate(predicted_prob = predict(m1, newdata = test, type = "response"))
```

Note that the predictions are in probability format. Convert these predictions into the categorical response variable using `ifelse()`. Note that the reference level is alphabetically first.

```r
train <- train |>
  mutate(predict_val = ifelse(predicted_prob > 0.5, "O", "B"))
```

Calculate the misclassification rate for each dataset.

```r
train <- train |>
  mutate(predict_val = ifelse(predicted_prob > 0.5, "O", "B"))
test <- test |>
  mutate(predict_val = ifelse(predicted_prob > 0.5, "O", "B"))
```

Section 5: LDA vs. QDA vs. Naive Bayes

Using the same training and testing split from above, use `lda()` to fit a model to predict species using frontal lobe size and carapace length and calculate the misclassification rate. How does this compare to logistic regression's misclassification rate?

The LDA model had a misclassification rate of 0.08125, which means it got about 8% of the test predictions wrong. Logistic regression also fit the data well, but based on the misclassification rate, LDA did slightly better at predicting species in this case.

```r
m_lda <- lda(sp~FL + CL, data = train)
train |>
  mutate(predicted_val = (predict(m_lda,
newdata = train)$class)) |>
  summarize(
    misclassification_rate = mean(sp !=
                                    predicted_val)
  )
```

```
##   misclassification_rate
## 1                0.08125
```

Repeat with qda().

QDA gave the same misclassification rate as LDA, meaning both models predicted species equally well on the test data.

```r
m_qda <- qda(sp~FL + CL, data = train)
train |>
  mutate(predicted_val = (predict(m_lda,
newdata = train)$class)) |>
  summarize(
    misclassification_rate = mean(sp !=
                                    predicted_val)
  )
```

```
##   misclassification_rate
## 1                0.08125
```

Repeat with naiveBayes().

Compared to LDA and QDA which had like an 8% misclassification error rate, Naive Bayes performed worse at predicting species with a 33% misclassification error rate. A likely reason is that Naive Bayes assumes the predictors are independent, which probably isn't true for FL and CL (they're likely correlated), making its assumptions less appropriate for this dataset.

```r
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.5.1
```

```r
m_nb <- naiveBayes(sp ~ FL + CL, data = train)

train |>
  mutate(predicted_val = predict(m_nb, newdata = train)) |>
  summarize(
    misclassification_rate = mean(sp != predicted_val)
  )
```

```
##   misclassification_rate
## 1                0.33125
```

Section 6: Sentiment Analysis

Load the `tidytext` library.

```r
library(tidytext)
```

```
## Warning: package 'tidytext' was built under R version 4.5.1
```

Load the `textdata` library.

```r
library(textdata)
```

```
## Warning: package 'textdata' was built under R version 4.5.1
```

Load the `janeaustenr` library.

```r
library(janeaustenr)
```

```
## Warning: package 'janeaustenr' was built under R version 4.5.1
```

The following code creates a data frame called `tidy_books` . What are the columns of the data set? And what do those columns represents.

Contains the column 'book', 'linenumber', 'chapter', and 'word' with the book column showing the name of the book. Linenumber is at what line the word was written, chapter being where each word was found, and the words column being each word written in the specific book.

```r
tidy_books <- austen_books() |>
  group_by(book) |>
  mutate(
    linenumber = row_number(),
    chapter = cumsum(str_detect(text,
                                regex("^chapter [\\divxlc]", ignore_case = TRUE)))) |>
  ungroup() |>
  unnest_tokens(word, text)
```

Use `unique()` to determine which different books are included in this dataset.

```
unique(tidy_books$book, 10)
```

```
## [1] Sense & Sensibility Pride & Prejudice   Mansfield Park
## [4] Emma                Northanger Abbey    Persuasion
## 6 Levels: Sense & Sensibility Pride & Prejudice Mansfield Park ... Persuasion
```

Use `filter()` to create a data set of only one of the books in the data set.

```
sense <- tidy_books |>
  filter(book == "Sense & Sensibility")
```

How many words are in this book?

```
nrow(sense)
```

```
## [1] 119957
```

Use the help file of `get_sentiments()` to pick a lexicon.

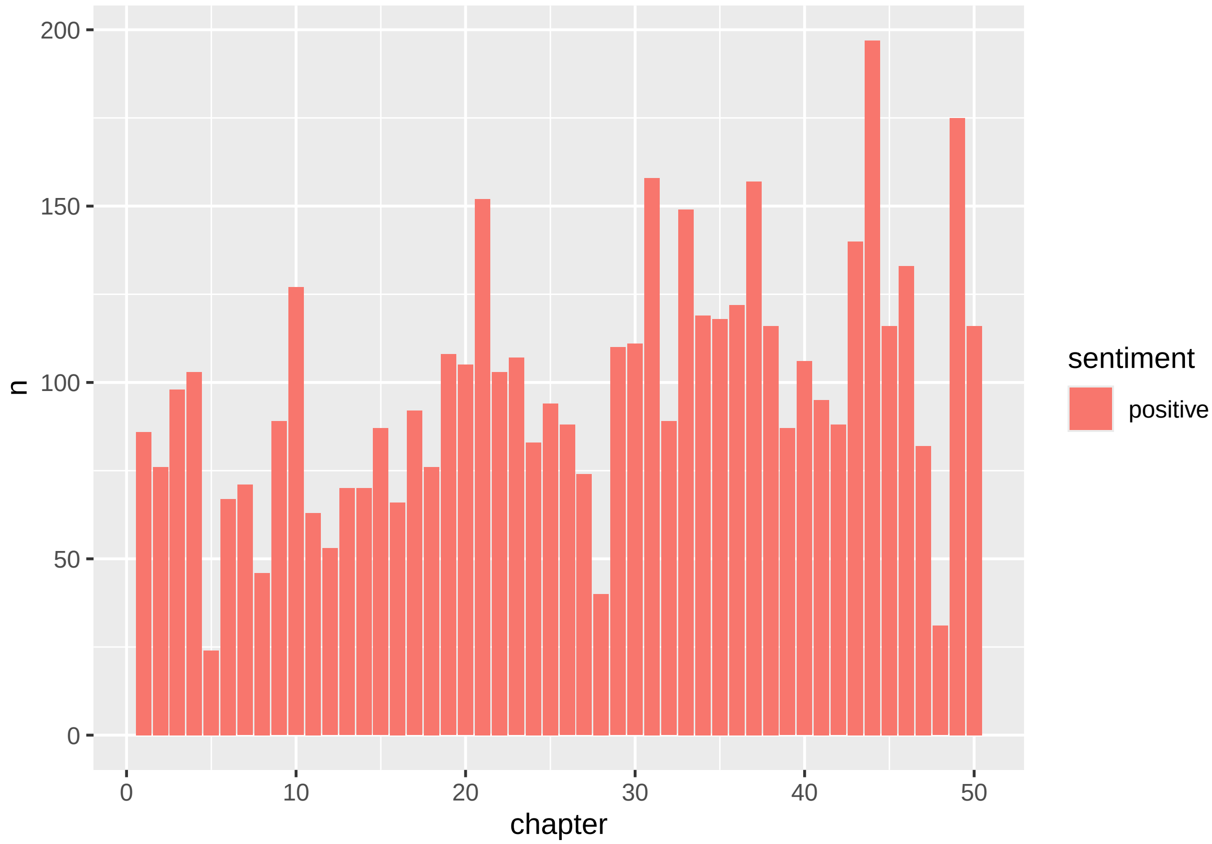
```
bing_sentiments <- get_sentiments("bing")
```

Pick one of the above lexicons and use `inner_join()` with your individual book dataset.

```
sense_sentiment <- sense |>
  inner_join(bing_sentiments, by = "word")
```

Filter for a specific sentiment and find the count for each chapter. Plot this using `geom_col()` .

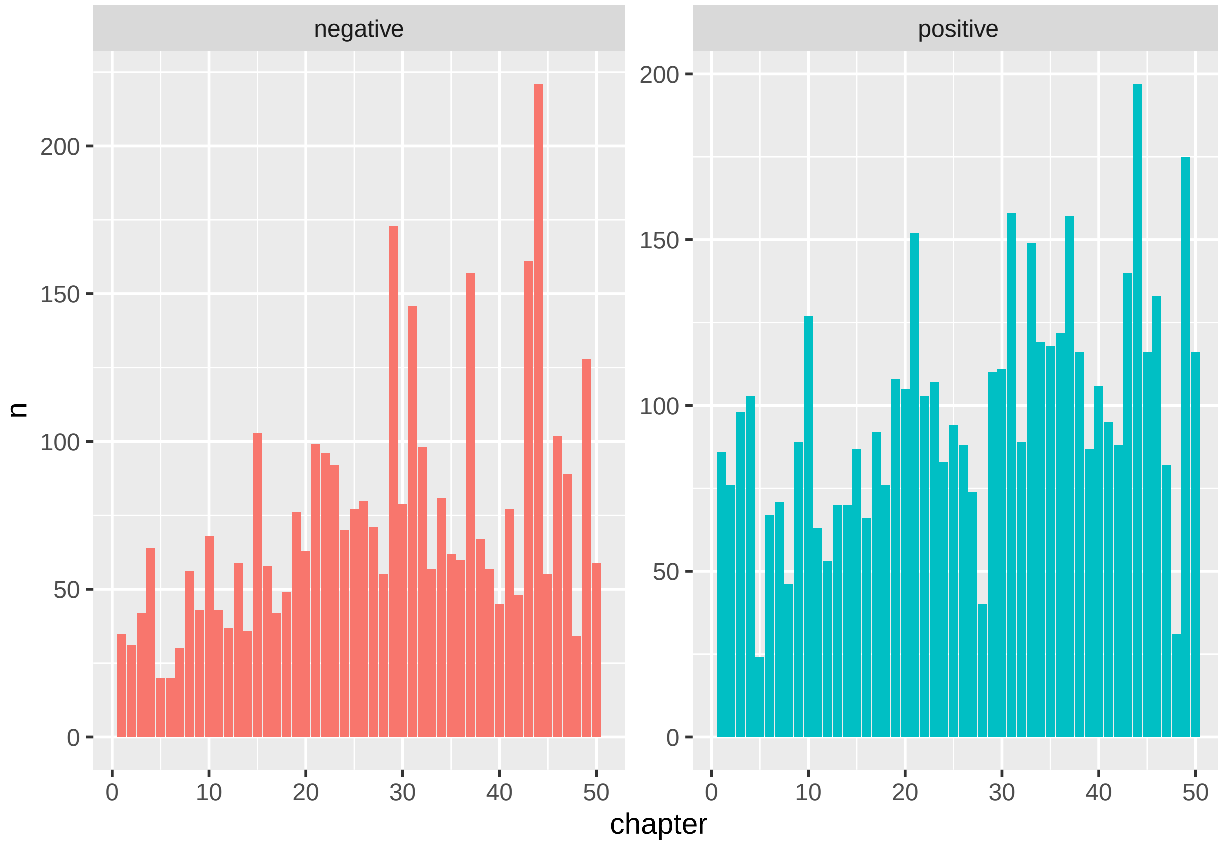
```
positive_by_chapter <- sense_sentiment |>
  filter(sentiment == "positive") |>
  count(chapter, sentiment)
ggplot(positive_by_chapter, aes(x = chapter, y = n, fill = sentiment)) +
  geom_col()
```



Create another graphic using this dataset using facet.

```
sentiment_by_chapter <- sense_sentiment |>
  count(chapter, sentiment)

ggplot(sentiment_by_chapter, aes(x = chapter, y = n, fill = sentiment)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~sentiment, scales = "free_y")
```


negative
positive
n
chapter
0
50
100
150
200
0
10
20
30
40
50